

Modular and Automorphic Functions

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Binary quadratic forms, quadratic fields, and Modular functions

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Part I: Motivation

A Fundamental Ambiguity

Theorem 1

The fundamental periods $2\omega_1$ and $2\omega_3$ of the Weierstrass function $\wp$ are not unique.

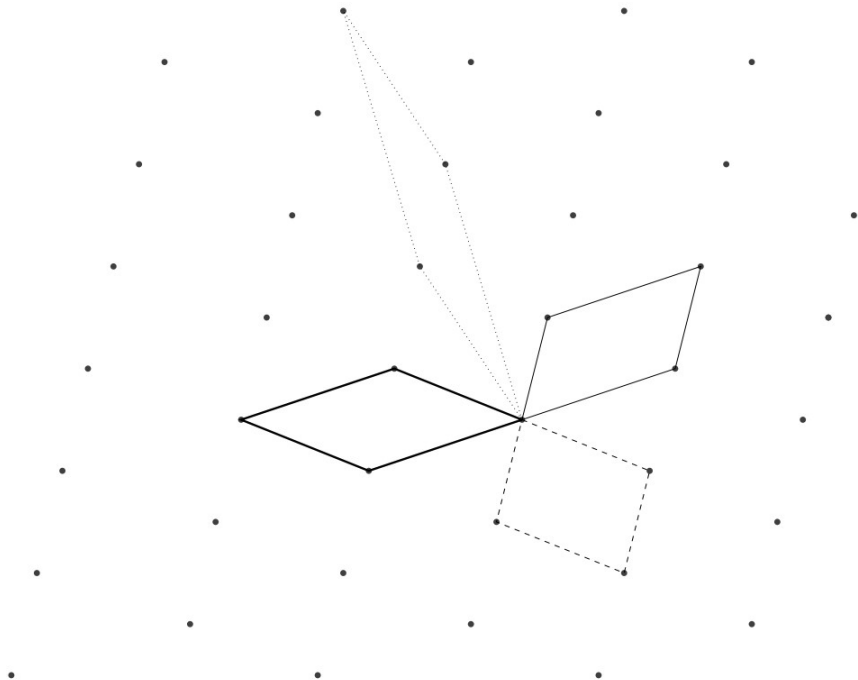
Indeed, any $2\omega'_1, 2\omega'_3$ related to the originals by

$$2\omega'_1 = 2a\omega_1 + 2b\omega_3,$$

$$2\omega'_3 = 2c\omega_1 + 2d\omega_3,$$

with $a, b, c, d \in \mathbb{Z}$ and $ad - bc = \pm 1$ yield the same period module:

A Fundamental Ambiguity



A Fundamental Ambiguity

Theorem 2

The periodicity of the Jacobian elliptic functions is completely determined by the ratio $\tau = \frac{\omega_3}{\omega_1}$.

Proof: We can use homogeneity relations!

$$\wp(tz; t\omega_1, t\omega_3) = t^{-2}\wp(z; \omega_1, \omega_3), \quad \zeta(tz; t\omega_1, t\omega_3) = t^{-1}\zeta(z; \omega_1, \omega_3), \\ \sigma(tz; t\omega_1, t\omega_3) = t\sigma(z; \omega_1, \omega_3) \quad \text{and} \quad \sigma_j(tz; t\omega_1, t\omega_3) = \sigma_j(z; \omega_1, \omega_3).$$

Then with $\mu(\omega_1, \omega_3) = \sqrt{e_1 - e_3}$, we can calculate, e.g.,

$$\operatorname{sn}(u; t\omega_1, t\omega_3) = \mu(t\omega_1, t\omega_3) \frac{\sigma(u/\mu(t\omega_1, t\omega_3); t\omega_1, t\omega_3)}{\sigma_3(u/\mu(t\omega_1, t\omega_3); t\omega_1, t\omega_3)} = \dots = \operatorname{sn}(u; \omega_1, \omega_3).$$

Now choose $t = \frac{1}{\omega_1}$!



A Fundamental Ambiguity

Our proof made use of the definition

$$\operatorname{sn}(u) = \sqrt{e_1 - e_3} \frac{\sigma(z)}{\sigma_3(z)}, \quad u = z\sqrt{e_1 - e_3} \in \mathbb{C},$$

so the related definitions

$$\operatorname{cn}(u) = \frac{\sigma_1(z)}{\sigma_3(z)}, \quad \operatorname{dn}(u) = \frac{\sigma_2(z)}{\sigma_3(z)}, \quad u = z\sqrt{e_1 - e_3} \in \mathbb{C}$$

can clearly extend our reasoning to the remaining Jacobi functions. Theorem 2 even holds for the **Jacobi tangent function**

$$\operatorname{tn}(u) = \frac{\operatorname{sn}(u)}{\operatorname{cn}(u)} = \sqrt{e_1 - e_3} \frac{\sigma(z)}{\sigma_1(z)}, \quad u = z\sqrt{e_1 - e_3} \in \mathbb{C},$$

another order-2 elliptic function, but with fundamental periods $2K$ and $2iK'$. (Recall that $K = \omega_1\sqrt{e_1 - e_3}$ and $iK' = \omega_3\sqrt{e_1 - e_3}$).

A Fundamental Ambiguity

Notice that Theorem 2 does not hold for the Weierstrass function! Recall that we're allowed transformations $(2\omega_1, 2\omega_3) \mapsto (2\omega'_1, 2\omega'_3)$ of the (unimodular) form

$$2\omega'_1 = 2a\omega_1 + 2b\omega_3,$$

$$2\omega'_3 = 2c\omega_1 + 2d\omega_3,$$

with $a, b, c, d \in \mathbb{Z}$ and $ad - bc = \pm 1$. Choose $a, d = 0$ and $b, c = 1$. This sends

$$(2\omega_1, 2\omega_3) \mapsto (2\omega_3, 2\omega_1) \quad \text{and} \quad \tau = \frac{\omega_3}{\omega_1} \mapsto \frac{\omega_1}{\omega_3} = \frac{1}{\tau},$$

but preserves the period lattice!

Preferred Periods

Which fundamental periods among our many choices are best?

Definition

We call the pair $(2\omega_1, 2\omega_3)$ of fundamental periods **reduced** if a) $2\omega_1$ has the smallest possible (nonzero) absolute value and b) $2\omega_3$ has the smallest possible absolute value subject to the constraint that $\text{Im}(\omega_3/\omega_1) > 0$.

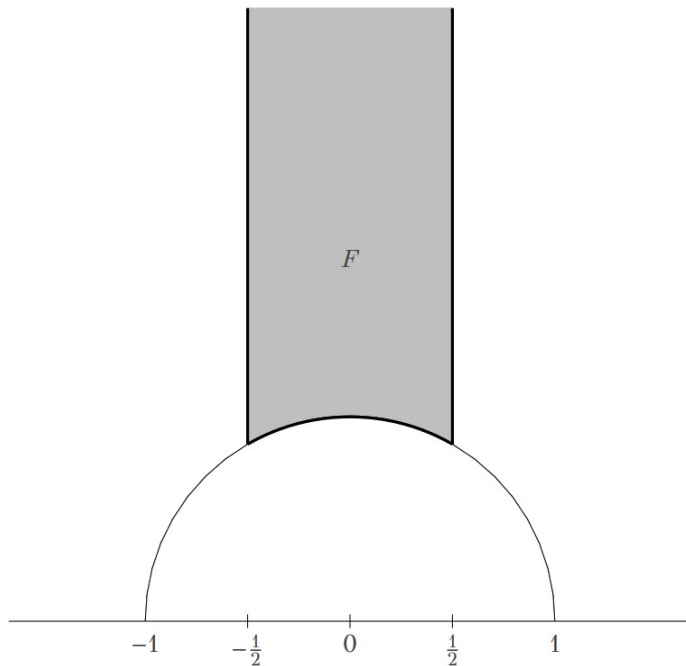
We can extend Theorem 2 to $\wp$ by requiring its periods to be reduced!

Theorem 3

The pair $(2\omega_1, 2\omega_3)$ of fundamental periods is reduced if and only if their ratio $\tau = \frac{\omega_3}{\omega_1}$ lies in the region

$$F = \left\{ \tau \in \mathbb{C} \mid |\tau| \geq 1, \text{Re}(\tau) \in \left[-\frac{1}{2}, \frac{1}{2} \right], \text{Im}(\tau) > 0 \right\}$$

Preferred Periods



Quantifying Ambiguity

We're now in position to discuss the symmetries we would like to observe.

Consider the following continuum:

The Trivial Group — — — — — Groups with transitive action on $\mathbb{C}$

Our groups should all be made of symmetries. Recall that

$$\text{Aut}(\mathbb{C}) = \{f(z) = az + b \mid a \in \mathbb{C} \setminus \{0\}, b \in \mathbb{C}\}.$$

Compactifying $\mathbb{C}$ into the Riemann sphere, a much richer structure emerges:

$$\text{Aut}(\mathbb{C}_\infty) = \left\{ T(z) = \frac{az + b}{cz + d} \mid a, b, c, d \in \mathbb{C}, ad - bc \neq 0 \right\}.$$

Quantifying Ambiguity

To uniquely associate a matrix representation to each Möbius transformation, we start with

$$\mathrm{GL}(2, \mathbb{C}) = \left\{ T = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid \det(T) = ad - bc \neq 0 \right\}$$

and quotient by the subspace spanned by the identity, resulting in

$$\begin{aligned} \mathrm{Aut}(\mathbb{C}_\infty) &\cong \mathrm{GL}(2, \mathbb{C}) / \{ \lambda I \mid \lambda \in \mathbb{C} \} = \mathrm{PSL}(2, \mathbb{C}) \\ &= \left\{ T = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \mathbb{C}, ad - bc = 1 \right\}. \end{aligned}$$

We can learn more about this group by considering its generators.

The Generators of $\text{Aut}(\mathbb{C}_\infty)$

The group $\text{PSL}(2, \mathbb{C})$ is generated by four (types of) transformations:

- ❶ Rotations R_θ given by $R_\theta = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & 1 \end{pmatrix}$ or $R_\theta(z) = e^{i\theta}z$, with $\theta \in \mathbb{R}$.
- ❷ Complex inversion given by $V = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$ or $V(z) = \frac{1}{z}$.
- ❸ Dilations centered at 0, given by $D_r = \begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix}$ or $D_r(z) = rz$, with $r > 0$.
- ❹ Translations given by $T_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ or $T_t(z) = z + t$ for $t \in \mathbb{C}$.

The Generators of $\text{Aut}(\mathbb{C}_+)$

Our primary concern will be for the open upper half-plane $\mathbb{C}_+$ - what are its symmetries? In a small miracle, $\text{Aut}(\mathbb{C}_+) \cong \text{PSL}(2, \mathbb{R})$, which is generated by

- ❶ Negative complex inversion given by $B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ or $B(z) = -\frac{1}{z}$.
- ❷ Dilations centered at 0, given by $D_r = \begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix}$ or $D_r(z) = rz$, with $r > 0$.
- ❸ Translations given by $T_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ or $T_t(z) = z + t$ for $t \in \mathbb{C}$.

Definition

We call a group **Fuchsian** if it is a discrete subgroup of $\text{PSL}(2, \mathbb{R})$.

The Modular Group

We introduce the **modular group**: the Fuchsian group $M = \mathrm{PSL}(2, \mathbb{Z})$.

This group has structure $M \cong \mathrm{SL}(2, \mathbb{Z}) / \{\pm I\}$, and is generated by:

- ❶ Negative complex inversion given by $B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ or $B(z) = -\frac{1}{z}$.
- ❷ Translation by 1 given by $T_1 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ or $T_1(z) = z + 1$.

Definition

A function f is called **modular** if it is meromorphic on $\mathbb{C}_+$ and satisfies

$$f(T(z)) = f(z), \quad z \in \mathbb{C}_+, T \in M.$$

Demystifying F

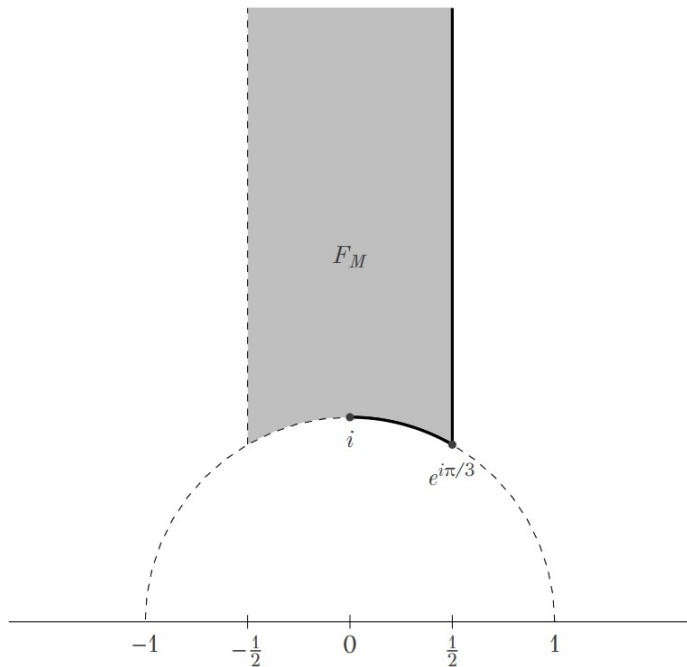
Recall the set

$$F = \left\{ \tau \in \mathbb{C} \mid |\tau| \geq 1, \operatorname{Re}(\tau) \in \left[-\frac{1}{2}, \frac{1}{2} \right], \operatorname{Im}(\tau) > 0 \right\}.$$

It is (almost) the **fundamental domain** of the modular group M . That is:

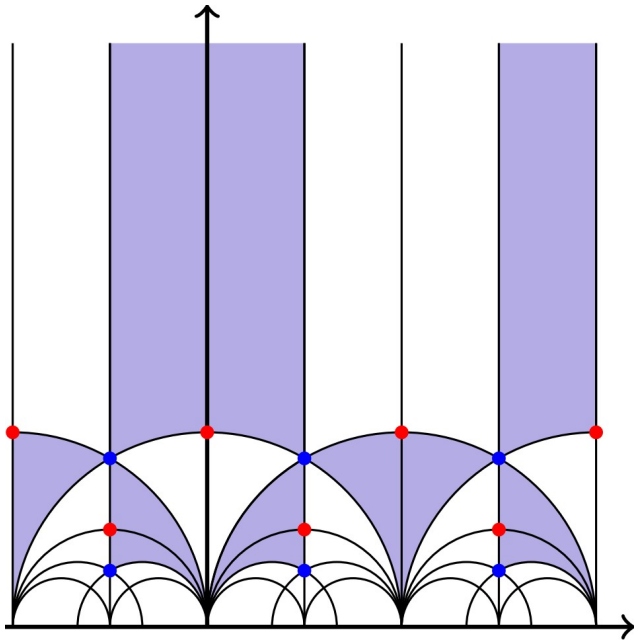
- 1 If $z \in \mathbb{C}_+ \setminus F$, then there exists some $z_M \in F$ and a $T \in M$ such that $T(z_M) = z$.
- 2 If z and z_M both lie in F , there is no $T \in M$ such that $T(z_M) = z$.

“The” Fundamental Domain F_M

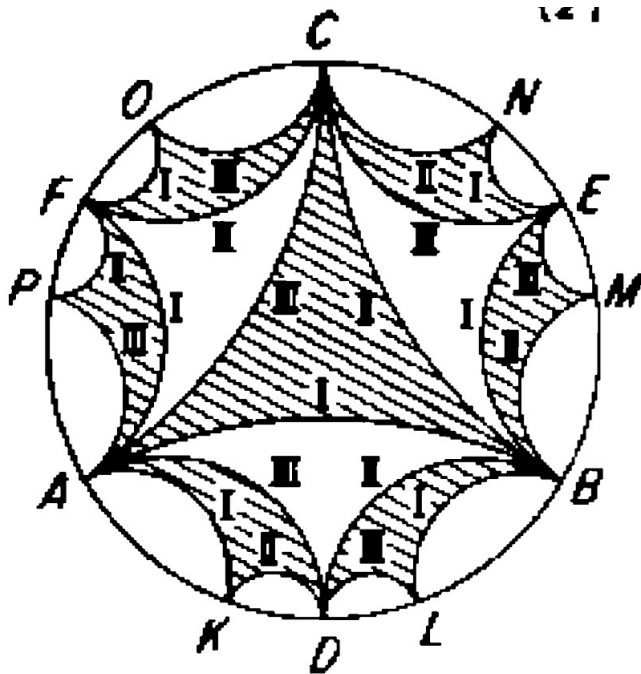


Demystifying F

Applying modular transformations to F tessellates $\mathbb{C}_+$:



A View from the Beltrami-Poincaré Disk



Our Flavor of Groups

The modular group is of the right flavor for our purposes. Recall our continuum:

The Trivial Group — — — — — Groups with transitive action on $\mathbb{C}$

The group M lies in the middle! Let's make this more precise:

Definition

A group G of Möbius transformations is **properly discontinuous** on $\mathbb{C}_+$ if there exists a point $z_0 \in \mathbb{C}_+$ such that the points $T(z_0)$ with $T \in G$ all lie outside some fixed neighborhood of z_0 .

Definition

The function f is **automorphic** with respect to some properly discontinuous group G on $\mathbb{C}_+$ if f is meromorphic on $\mathbb{C}_+$ and satisfies

$$f(T(z)) = f(z), \quad z \in \mathbb{C}_+, T \in G.$$

There are no nonconstant functions which are automorphic with respect to **continuous groups** or **improperly discontinuous groups**.

Facts About Automorphicity

Propositions

- It suffices to verify automorphicity in a neighborhood.
- Any two simple functions automorphic with respect to the same group have an algebraic relation.
- Derivatives of automorphic functions are not usually automorphic...
- If f is simple and automorphic, then so is

$$\frac{2f'f''' - 3(f'')^2}{2(f')^4} = \frac{D_S f}{(f')^2}.$$

Aside: The Schwarzian Derivative D_S

The formal operator

$$D_S f = \frac{2f'f''' - 3(f'')^2}{2(f')^2} = (\ln(f'))'' - \frac{1}{2} \left((\ln(f'))' \right)^2$$

was introduced by H. A. Schwarz, and has $\ker(D_S) = \{\text{Möbius transformations}\}$. Its usefulness in our context stems from its chain rule

$$D_S[f \circ g] = ((D_S f) \circ g)(g')^2 + D_S g$$

in cases where g is a Möbius transformation and f is automorphic.

One remarkable outworking of D_S 's role in automorphy preservation is:

Quotient Representation Theorem

If w is a non-constant simple automorphic function of z , then z can be expressed as a function of w by the quotient of two solutions of a second-order linear differential equation of the form

$$\frac{d^2 \eta}{dw^2} = u \eta,$$

where u is an algebraic function of w .

Part II: Examples

The Function J

Let's develop a modular function! The Weierstrass invariants satisfy

$$g_2(\omega_1, \omega_3) = 60 \sum' \frac{1}{\Omega^4} = \frac{1}{\omega_1^4} g_2(1, \tau),$$

$$g_3(\omega_1, \omega_3) = 140 \sum' \frac{1}{\Omega^6} = \frac{1}{\omega_1^6} g_3(1, \tau),$$

so the discriminant has

$$D(\omega_1, \omega_3) = \frac{1}{16} (g_2^3(\omega_1, \omega_3) - 27g_3^2(\omega_1, \omega_3)) = \frac{1}{\omega_1^{12}} D(1, \tau).$$

To obtain a true modular function, we must eliminate the factor of ω_1^{-12} .

This is achieved by **Klein's absolute invariant**

$$J(\tau) = \frac{16g_2^3(\tau)}{D(\tau)}, \quad \tau \in \mathbb{C}_+.$$

Properties of J

Several attributes will interest us - one is the series

$$J(\tau) = \frac{1}{1728q^2} \left(1 + \sum_{n \in \mathbb{N}} c_n q^{2n} \right), \quad \tau \in \mathbb{C}_+,$$

written in terms of the nome $q = e^{i\pi\tau}$, where $c_n \in \mathbb{N}_0$.

Theorem 4

Let F_M be F with “half” of its boundary deleted. Then,

- If $a \in \mathbb{C}_\infty$, the equation $J(\tau) = a$ has exactly one root in F_M , save a double root at $\tau = i$ and a triple root at the corner $\tau = e^{i\pi/3}$ or $\tau = e^{2\pi i/3}$.
- If $a \in \mathbb{C} \setminus \mathbb{R}$, then $J(\tau) = a$ has exactly one root in F_M° .
- If $J(\tau) = J(\tau')$, then there exists a unique $T \in M$ such that $\tau' = T(\tau)$.
- Corollary: J never repeats values in F_M .
- The most general **simple** modular function is a rational function of J .

Simple Automorphic Functions

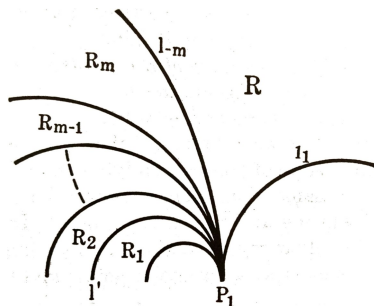
Definition

A function f which is automorphic with respect to G is called **simple** if:

- ① (f has no essential singularities at ordinary points* of G),
- ① The fundamental domain of G is a (possibly curvilinear) polygon,
- ② The limit $\lim_{z \rightarrow P} f(z)$ exists at each parabolic point** P of the fundamental domain.

(*: L. Ford, *Automorphic Functions*, p.8-42, **: Ford, p.59-64.)

A sample parabolic point P_1 :



Application: Finding Fundamental Periods

Weierstrass Inversion Theorem

Let g'' and g''' be complex numbers such that $D = \frac{1}{16}((g'')^3 - 27(g''')^2) \neq 0$. Then there exist ω_1, ω_3 with $\text{Im}(\omega_3/\omega_1) > 0$ such that the function $\wp(\cdot; \omega_1, \omega_3)$ satisfies the differential equation

$$(\wp')^2 = 4\wp^3 - g''\wp - g'''.$$

Proof: [Hurwitz-Courant p.227-228, Translated by Hunter Morris]

Our goal is to solve the equations

$$\begin{cases} g_2(\omega_1, \omega_3) := 60 \sum' \frac{1}{\Omega^4} = g'', \\ g_3(\omega_1, \omega_3) := 140 \sum' \frac{1}{\Omega^6} = g''' \end{cases}$$

for ω_1 and ω_3 .

Application: Finding Fundamental Periods

Case 1: Say $g'' = 0$. Introducing $\rho = e^{2\pi i/3}$, the equation $g_2(\omega_1, \rho\omega_1) = 0$ holds for any nonzero ω_1 . Then the choice

$$\omega_1^6 = \frac{140}{g'''} \sum' \frac{1}{(m_1 + m_3\rho)^6}$$

also satisfies $g_3(\omega_1, \rho\omega_1) = g'''$.

Case 2: Say $g''' = 0$. The equation $g_3(\omega_1, i\omega_1) = 0$ holds for any nonzero ω_1 . Then the choice

$$\omega_1^4 = \frac{60}{g''} \sum' \frac{1}{(m_1 + m_3i)^4}$$

also satisfies $g_2(\omega_1, i\omega_1) = g''$.

Application: Finding Fundamental Periods

Case 3 (generic): Now let g'' and g''' both be nonzero. We achieve our goal if and only if

$$\frac{g_2(\omega_1, \omega_3)}{g_3(\omega_1, \omega_3)} = \frac{g''}{g'''} \quad \text{and} \quad \frac{16g_2^3(\omega_1, \omega_3)}{g_2^3(\omega_1, \omega_3) - 27g_3^2(\omega_1, \omega_3)} = \frac{16(g'')^3}{D}$$

hold, which allow us to consider ω_1 and $\tau = \omega_3/\omega_1$ as the unknowns. In this light, our new pair of equations can be written

$$\omega_1^2 = \frac{g''}{g'''} \frac{\frac{140}{g'''} \sum' (m_1 + m_3 \tau)^{-6}}{\frac{60}{g''} \sum' (m_1 + m_3 \tau)^{-4}} \quad \text{and} \quad J(\tau) = \frac{16(g'')^3}{D}.$$

The second equation is solvable by the properties of J we have seen, and we can take square roots at left to find ω_1 . Together, these determine $\omega_3 = \tau\omega_1$.



Arithmetic Properties of J

The conspicuous 1728 in J 's Laurent series leads to frequent study of $j = 1728J$.

T. Schneider (1937) proved: $j(\tau)$ is an **algebraic integer** (a root of a monic polynomial with integer coefficients) if τ lies in the upper-half plane portion of the **quadratic imaginary field**. Such values $j(\tau)$ are called **singular moduli**.

Some examples:

- $j(i) = 1728$ (so $J(i) = 1!$),
- $j(i\sqrt{2}) = 1728 \left(\frac{5}{3}\right)^3 = 8000$,
- $j\left(\frac{1-i\sqrt{3}}{2}\right) = 0$,
- $j\left(\frac{1+i\sqrt{5}}{2}\right) = \frac{1}{2}(-191025 - 85995\sqrt{5})$,
- $j\left(\frac{-1+\sqrt{-163}}{2}\right) = -262537412680768000 = -(640320)^2$.

Notice that

$$\begin{aligned} j\left(\frac{-1 + \sqrt{-163}}{2}\right) &= -262537412680768000 \\ &\approx -262537412640768000 = -e^{\pi\sqrt{163}} + 744. \end{aligned}$$

Arithmetic Properties of J

Transcendence Theorems

- C. Siegel (1932): Fix $g'', g''' \in \mathbb{C}$ such that $D = \frac{1}{16}((g'')^3 - 27(g''')^2) \neq 0$, and let $\wp = \wp(\cdot; g'', g''')$ be the resulting Weierstrass function. If both g'' and g''' are algebraic numbers, then at least one of the fundamental periods of $\wp$ is transcendental.
- T. Schneider (1937):
 - i) Under Siegel's assumptions, all periods of $\wp$ are transcendental.
 - ii) If $\tau \in \mathbb{C}_+$ is not algebraic, but a member of the quadratic imaginary field, then $j(\tau)$ is transcendental.
- Y. Nesterenko (1996): Let $\tau \in \mathbb{C}_+$ have $e^{2\pi i\tau}$ algebraic. Then the three numbers $j(\tau)$, $\frac{j'(\tau)}{\pi}$, and $\frac{j''(\tau)}{\pi^2}$ are algebraically independent.

Another Fuchsian Group

To define a “genuinely” automorphic function, we need a smaller Fuchsian group. Let M_2 be the subgroup of M generated by:

① Translation by 2 given by $T_2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ or $T_2(z) = z + 2$.

② $BT_2B = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$ or $BT_2B(z) = \frac{z}{1-2z}$.

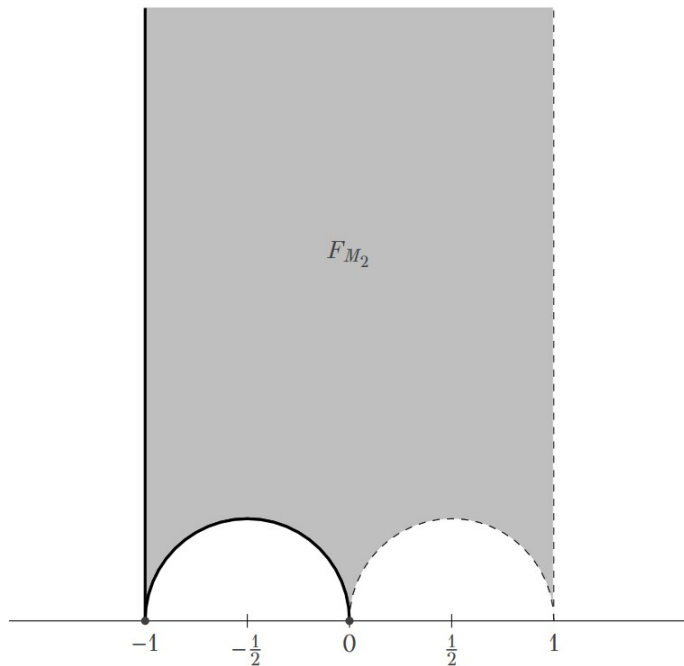
Then M_2 is a Fuchsian group which can be written

$$M_2 = \left\{ T = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \mathbb{Z}, a, d \text{ odd}, b, c \text{ even}, ad - bc = 1 \right\}$$

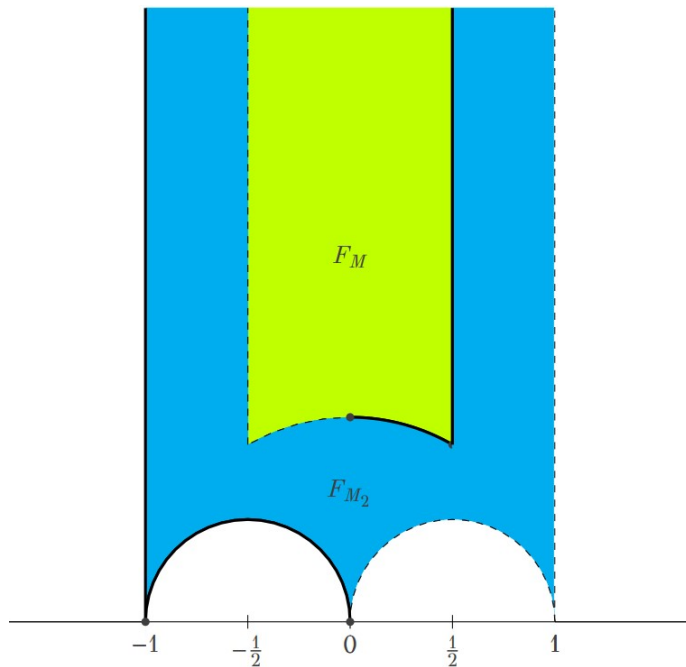
with fundamental domain

$$F_{M_2} = \left\{ z \in \mathbb{C}_+ \mid \operatorname{Re}(z) \in [-1, 1], \left| z + \frac{1}{2} \right| \geq \frac{1}{2}, \left| z - \frac{1}{2} \right| > \frac{1}{2} \right\}.$$

The Fundamental Domain F_{M_2}



A Comparison



A Genuinely Automorphic Function

With these notions, we can define the **modular λ function**

$$\lambda(\tau) = k^2(\tau) = \frac{e_2 - e_3}{e_1 - e_3}, \quad \tau \in \mathbb{C}_+,$$

which is automorphic with respect to M_2 .

Theorem 5

- Let $a \in \mathbb{C}_\infty \setminus \{0\}$. Then $\lambda(\tau) = a$ has exactly one root in F_{M_2} .
- We can recover J from λ via

$$J = \frac{4}{27} \frac{(1 - \lambda + \lambda^2)^3}{\lambda^2(1 - \lambda)^2}.$$

- The most general simple automorphic function (with respect to M_2) is a rational function of λ

General Automorphic Functions

Both J and λ can rationally represent all other simple functions which are automorphic with respect to the same group! Why?

Theorem 6 (Ford, p.97)

Given a properly discontinuous group G , if there exists a function f_1 which is automorphic with respect to G and has only one pole in the fundamental domain, then any simple function automorphic with respect to G which shares f_1 's domain can be written as a rational function of f_1 .

Part III: Modular Forms and Beyond

A New Motivation

We were fortunate to already have encountered modular and automorphic functions - we had only to recognize echoes this property in quantities like g_2 and g_3 . What if we had discovered the Fuchsian groups before these functions?

H. Poincaré (1882) considered such questions, first for *finite* Fuchsian groups. Given a rational function R , and a finite Fuchsian group $G_N = (T_n)_0^N$ with $T_0 = I$, he constructed the series

$$P_{G_N}(\tau) = \tau + R(T_1(\tau)) + R(T_2(\tau)) + \dots = \sum_{n=0}^N R(T_n(\tau)), \quad \tau \in \mathbb{C}_+.$$

The function P_{G_N} is automorphic with respect to G_N .

Anharmonic (Cross) Ratios

One finite Fuchsian group consists of the following six transformations:

$$I(\tau) = \tau, \quad V(\tau) = \frac{1}{\tau}, \quad T_-(\tau) = 1 - \tau, \\ VT_-(\tau) = \frac{1}{1 - \tau}, \quad T_-V(\tau) = \frac{\tau - 1}{\tau}, \quad \text{and} \quad VT_-V(\tau) = \frac{\tau}{\tau - 1}, \quad \tau \in \mathbb{C}_+.$$

A New Motivation

For infinite Fuchsian groups $G_\infty = (T_n)_0^\infty$, the series P_{G_∞} may not converge, but the **Poincaré θ -series**

$$\theta_{k,G_\infty}(\tau) = \sum_{n \in \mathbb{N}_0} (c_n \tau + d_n)^{-2k} R(T_n(\tau)), \quad \tau \in \mathbb{C}_+, k \in \mathbb{N}$$

does when $k \geq 2$. It is no longer automorphic with respect to the underlying group, but instead transforms according to

$$\theta_{k,G_\infty}(T_n(\tau)) = (c_n \tau + d_n)^{2k} \theta_{k,G_\infty}(\tau), \quad n \in \mathbb{N}_0, \tau \in \mathbb{C}_+.$$

Quotients of such series can be used to manufacture functions which are automorphic with respect to G_∞ .

Relaxing Automorphicity

There are many functions which fail to be automorphic in a well-structured way, like the θ -series.

Definition

A meromorphic function f on $\mathbb{C}_+$ is called a **modular form** of weight $k \geq 0$ for the (properly discontinuous) group G if

$$f(T(\tau)) = (c\tau + d)^k f(\tau), \quad z \in \mathbb{C}_+, T = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in G.$$

(Technically, we should call these **automorphic forms** if G isn't the modular group, but this distinction isn't made in the literature).

Nearly all of our functions so far meet this definition - to see this, let w denote the weight of a modular form with respect to the modular group $M = \mathrm{PSL}(2, \mathbb{Z})$.

Examples of Modular Forms

- The Weierstrass invariants have $w(g_2) = 4$ and $w(g_3) = 6$.
- The **Eisenstein series** $Es_k(\omega_1, \omega_3) = \sum' \Omega^{-2k}$ have $w(Es_k) = 2k$.
- The Discriminant $D = \frac{1}{16}(g_2^3 - 27g_3^2)$ has $w(D) = 12$.
- The q -series

$$\sum_{k \in \mathbb{Z}} e^{2\pi i k^2 \tau} = \sum_{k \in \mathbb{Z}} q^{2k^2} = \vartheta_3(0; \tau)$$

has weight $w(\vartheta_3(0; \cdot)) = \frac{1}{2}$.

- The **Dedekind η function**

$$\eta(\tau) = q^{1/12} \prod_{k \in \mathbb{N}} (1 - q^{2k}), \quad \tau \in \mathbb{C}_+$$

has $w(\eta) = \frac{1}{2}$.

- Klein's absolute invariant has $w(J) = 0$ (it's a modular function).
- The automorphic function λ has weight 0 with respect to M_2 .

Examples of Modular Forms

Since all modular forms are 1-periodic (after all, the translation-by-one map T_1 has $c = 0$ and $d = 1$), a generic modular form f has the Fourier expansion

$$f(\tau) = \sum_{n=n_0}^{\infty} a_n e^{2\pi i n \tau} = \sum_{n=n_0}^{\infty} a_n q^{2n}$$

for some $n_0 \in \mathbb{N}$.

Definition

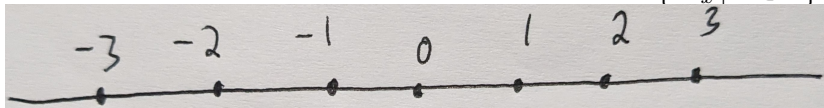
A modular form f is called a **cusp form** if it has $n_0 > 0$.

With this notion, and recalling that $Es_k(\omega_1, \omega_3) = \sum' \Omega^{-2k}$, we can establish a few facts:

- The spaces of modular forms of weight 4, 6, 8, and 10 are spanned by Es_2 , Es_3 , Es_2^2 , and Es_2Es_3 , respectively.
- The weight-12 space is spanned by Es_2^3 and Es_3^2 .
- The discriminant D is a lowest-weight cusp form, and in the normalization $\tilde{D} = \frac{1}{1728}(Es_2^3 - Es_3^2)$ has $\tilde{D} = \eta^{12}$.

Lattices and Duals

The one-dimensional lattice $\mathbb{Z}$ can be viewed as the set $\Lambda = \{k\hat{e}_x \mid k \in \mathbb{Z}\}$.



To *any* lattice Λ in $\mathbb{R}^n$, we can associate a **dual lattice**

$\Lambda^* = \{\lambda^* \in \mathbb{R}^n \mid \forall \lambda \in \Lambda [\langle \lambda^*, \lambda \rangle \in \mathbb{Z}]\}$ where for the cases considered here, $\langle \cdot, \cdot \rangle$ is the inner (dot) product of $\mathbb{R}^n$. $\langle \cdot, \cdot \rangle$ has **positive definite** signature.

For $\Lambda = \mathbb{Z}$, ($n = 1$), $\Lambda^* = \mathbb{Z}$ since for each $\lambda^* = k^*\hat{e}_x \in \Lambda^*$, $k^*k \in \mathbb{Z} \forall k \in \mathbb{Z}$

Definition

If $\Lambda = \Lambda^*$ we call the lattice Λ **self-dual** (or **unimodular**)

The **Theta function** Θ_Λ associated to a lattice Λ with bilinear form $\langle \cdot, \cdot \rangle$ is

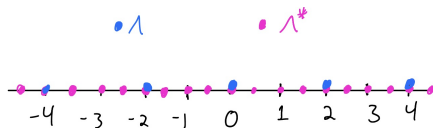
$$\Theta_\Lambda = \sum_{\lambda \in \Lambda} q^{\langle \lambda, \lambda \rangle}$$

Using Poisson summation, we can see that

$$\Theta_{\mathbb{Z}}(\tau) = \sum_{k \in \mathbb{Z}} e^{\pi i k^2 \tau} = \sqrt{\frac{i}{\tau}} \sum_{k \in \mathbb{Z}} e^{-\pi i k^2 / \tau} = \sqrt{\frac{i}{\tau}} \Theta_{\mathbb{Z}}\left(-\frac{1}{\tau}\right), \quad \tau \in \mathbb{C}_+.$$

More One-Dimensional Lattices

For $\Lambda = 2\mathbb{Z}$, each $\lambda^* \in \Lambda^*$ must satisfy $2m\lambda^* \in \mathbb{Z}$ for all $m \in \mathbb{Z}$, so therefore $\Lambda^* = \frac{1}{2}\mathbb{Z}$



Definition

We say the lattice Λ is **even** if $\langle \lambda, \lambda \rangle$ is even for each $\lambda \in \Lambda$.

Notice that

$$\Theta_{2\mathbb{Z}}(\tau) = \sum_{\lambda \in 2\mathbb{Z}} q^{\langle \lambda, \lambda \rangle} = \sum_{k \in \mathbb{Z}} q^{4k^2} = \sum_{k \in \mathbb{Z}} e^{4\pi i k^2 \tau} = \vartheta_3(0; 2\tau), \quad \tau \in \mathbb{C}_+.$$

This lattice $\Lambda = 2\mathbb{Z}$ is even, but not self-dual. Conversely, $\Lambda = \mathbb{Z}$ is self-dual, but not even. These properties are manifested by the associated Theta functions:

$$\Lambda \text{ self-dual} \Rightarrow \Theta_{\Lambda}\left(-\frac{1}{\tau}\right) = \left(\frac{\tau}{i}\right)^{n/2} \Theta_{\Lambda}(\tau) \quad \Lambda \text{ even} \Rightarrow \Theta_{\Lambda}(\tau) = \Theta_{\Lambda}(\tau + 1)$$

If a Λ is both even and self-dual, Θ_{Λ} is a modular form!

A Dimensional Constraint

Theorem 7

Even, self-dual lattices exist only in dimensions $n = 8k$, $k \in \mathbb{N}$.

Proof Sketch: Recall the generators $B(\tau) = -\frac{1}{\tau}$ and $T_1(\tau) = \tau + 1$, $\tau \in \mathbb{C}_+$ of the modular group. We know that $B(i) = i$, and can show that $T_1 B T_1 B T_1(i) = i$ also. Then we compute (using our observations about evenness and self-duality)

$$\begin{aligned}\Theta_{\Lambda}(i) &= \Theta_{\Lambda}(i+1) = \left(\frac{i+1}{i}\right)^{n/2} \Theta_{\Lambda}\left(-\frac{1}{1+i}\right) = (1-i)^{n/2} \Theta_{\Lambda}\left(\frac{i-1}{2}\right) \\ &= (1-i)^{n/2} \Theta_{\Lambda}\left(\frac{i+1}{2}\right) = \left(\frac{i+1}{2i}(1-i)\right)^{n/2} \Theta_{\Lambda}\left(-\frac{2}{i+1}\right) \\ &= (-i)^{n/2} \Theta_{\Lambda}(i-1) = (-i)^{n/2} \Theta_{\Lambda}(i).\end{aligned}$$

This implies that $n = 8k$ for some $k \in \mathbb{N}$.

□

Given such a dimension, how many even, self-dual lattices exist?

Positive-Definite, Even, Self-Dual Lattices

n	Count	Types
8	1	E_8 (sporadic Lie group)
16	2	$E_8 \times E_8$ and $SO(32)$
24	24	Niemeier Lattices (related to Moonshine)
32	11×10^8	

Bounds are given by the Smith-Minkowski-Siegel mass formula.

Example: The E_8 root lattice consists of the vectors of $\mathbb{R}^8$ whose components on a fixed orthonormal basis have coordinates which are either **all integer** or **all half integer** and which sum to an **even integer**. Its associated Theta function is

$$\Theta_{E_8}(\tau) = \sum_{\lambda \in E_8} q^{\lambda^2} = \frac{1}{2}(\vartheta_2^8(\tau) + \vartheta_3^8(\tau) + \vartheta_4^8(\tau)) = Es_4(\tau) = 1 + \textcolor{red}{240}q^2 + \dots$$

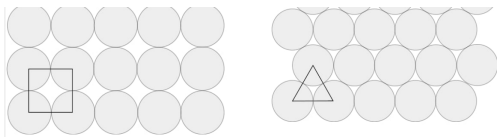
and satisfies

$$j(\tau)^{1/3} = \frac{\Theta_{E_8}(\tau)}{\eta(\tau)^8} = q^{-2/3}(1 + 248q^2 + 4124q^4 + 34752q^6 \dots)$$

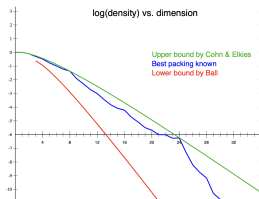
The dimensions of the three smallest E_8 irreps are: $1 + \textcolor{red}{248} + \textcolor{red}{3875} = \textcolor{red}{4124}$

Leech Lattice

The Leech Lattice is special among the 24 Niemeier lattices, as it has no root vectors (i.e. $\nexists \lambda \in \Lambda_{\text{Leech}}$ s.t. $\langle \lambda, \lambda \rangle = 2$). Related to the *Monster group*. Spheres with centers on its lattice points have the densest packing in dim. $n = 24$. In dimension 2, the **kissing number** (number of nearest neighbors to a given sphere), is 6 and therefore the associated automorphism group has order 12.



The E_8 lattice is the densest packing in dimension 8, with kissing number **240**.



$$\Theta_{\text{Leech}}(\tau) = 1 + \mathbf{196560}q^2 + 16773120q^3 + 398034000q^4 + \dots$$

The Periodic Table Of Finite Simple Groups

[†]Finite simple groups are determined by their order with the following exceptions:
 $B_n(q)$ and $C_n(q)$ for q odd, $n \geq 2$;
 $A_6 \cong A_2(2)$ and $A_1(4)$ of order 20160.

Group Theory Preliminaries

A **finite** group, is a group with a finite underlying set. A **simple** group has no **normal** subgroups. A subgroup N of a group G is **normal** if $gng^{-1} \in N$ for all $n \in N$ and $g \in G$.

Definition

Let G be a group, and H a (not-necessarily-normal) subgroup of G . The **left cosets** of H in G are the sets $gH = \{gh \mid h \in H\}$ for $g \in G$, and the **right cosets** of H in G are the sets $Hg = \{hg \mid h \in H\}$.

Denote the collection of left (respectively, right) H -cosets by: G/H (resp., $H \setminus G$). These sets are groups if H is **normal** in G .

Example: Let $G = (\mathbb{Z}, +)$, fix $m \in \mathbb{Z}$, and let $H = m\mathbb{Z}$. Then the distinct left $m\mathbb{Z}$ -cosets are

$$0 + H = \{\dots, -2m + 0, -m + 0, 0 + 0, m + 0, 2m + 0, \dots\},$$

$$1 + H = \{\dots, -2m + 1, -m + 1, 0 + 1, m + 1, 2m + 1, \dots\},$$

$$\vdots$$

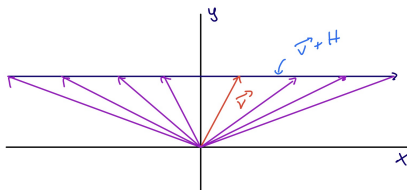
$$(m - 1) + H = \{\dots, -m + (m - 1), 0 + (m - 1), m + (m - 1), \dots\}.$$

Therefore, $G/H = \mathbb{Z}/m\mathbb{Z} = \{0, 1, 2, \dots, (m - 1)\}$.

More Coset Examples

Let $G = \mathbb{R}^2$, $H = \mathbb{R}$. We can interpret G as the set $\{(x, y)\}$ of 2-vectors and $H = \mathbb{R}_x$ as the set of x -directed vectors. These sets are additive groups.

Consideration of the H -cosets leads to the conclusion $G/H = \mathbb{R}^2/\mathbb{R}_x = \mathbb{R}_y$



Recall that $SL(2, \mathbb{R}) = \text{Aut}(\mathbb{C}_+)$.

For $\tau = x + iy \in \mathbb{C}_+$, introduce $Q_\tau = \begin{pmatrix} y^{-1/2} & -xy^{-1/2} \\ 0 & y^{1/2} \end{pmatrix} \in SL(2, \mathbb{R})$. By definition, Q_τ sends τ to i , and is invertible, so also $Q_\tau^{-1}(i) = \tau$.

$SO(2, \mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \mid a, b \in \mathbb{R}, a^2 + b^2 = 1 \right\}$. Notice that $SO(2, \mathbb{R})$ **stablizes**

i . i.e. $T(i) = i$ for all $T \in SO(2, \mathbb{R})$. Therefore there exists a 1-1 map from the coset (not quotient!) space $SL(2, \mathbb{R})/SO(2, \mathbb{R})$ into $\mathbb{C}_+$, **given by**:

$$Q_\tau^{-1}(i) = \begin{pmatrix} y^{1/2} & xy^{-1/2} \\ 0 & y^{-1/2} \end{pmatrix} (i) = \{yi + x \mid y > 0, x \in \mathbb{R}, y \in \mathbb{R}^+\}$$

Comments on Finite Simple Groups

Simple groups are like prime numbers, or building blocks out of which other groups are formed.

There are 18 infinite families of finite, simple groups (e.g. $\mathbb{Z}/p\mathbb{Z}$ for primes p), and also 26 “sporadic” groups, which do not belong to any family.

The largest sporadic group is the **Monster group** $\mathcal{M}$, which has order

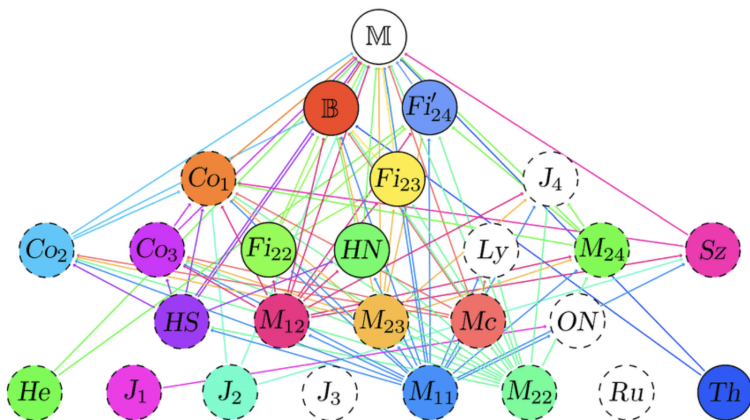
$$|\mathcal{M}| = 2^{46} \cdot 3^{20} \cdot 5^9 \cdot 7^6 \cdot 11^2 \cdot 13^3 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 41 \cdot 47 \cdot 59 \cdot 71 \approx 10^{53}$$

20 of the 26 sporadic groups are **subquotients** of $\mathcal{M}$, meaning they are of the form H/N for a subgroup H of $\mathcal{M}$ and a normal subgroup N of H .

These 20 are called the **happy family**

The other 6 known as **pariah** groups (or outcasts)

An Algebraic Interlude



Here, $\mathbb{M} = \mathcal{M}$, and directed lines illustrate subquotient relationships. For groups G with solid boundary circles, there is a known vector space V_G with $\text{Aut}(V_G) = G$.

A Pinch of Moonshine

Number theorists (like Imamoglu) work with $j = 1728J$, whose series has the form

$$j(\tau) = \frac{1}{q^2} + 744 + 196884q^2 + 21493760q^4 + \dots = \frac{1}{q^2} + \sum_{n \in \mathbb{N}_0} c'_n q^{2n},$$

where the c'_n grow like $\frac{e^{4\pi\sqrt{n}}}{\sqrt{2} n^{3/4}}$ as $n \rightarrow \infty$ (they're OEIS A001379). Notice that

$$c'_1 = 196884 = 196883 + 1,$$

$$c'_2 = 21493760 = 21296876 + 196883 + 1,$$

$$c'_3 = \dots$$

The **blue numbers** are dimensions of irreducible representations of $\mathcal{M}$!

The connection between $\mathcal{M}$ and modular functions runs much deeper, and goes by the name **moonshine** (i.e. "**foolish talk!**").

Observed by J. McKay (1978), formulated by J. Conway and S. Norton (1979), and proven by R. Borcherds (1992).

Representation Theory

A **representation** of a group G on a vector space V is a homeomorphism

$$\rho : G \rightarrow \mathrm{GL}(V).$$

A representation ρ is **irreducible** (an irrep.) if it has no sub-representations, ρ_1, ρ_2 :

$$\rho(g) = \begin{pmatrix} \rho_1(g) & 0 \\ 0 & \rho_2(g) \end{pmatrix} = \rho_1(g) \oplus \rho_2(g), \quad g \in G.$$

The **character** of the element $g \in G$ in a representation ρ is given by: $\chi_\rho(g) = \mathrm{tr}(\rho(g))$. If ρ is reducible into ρ_1 and ρ_2 , then

$$\chi_\rho(g) = \mathrm{tr}(\rho(g)) = \mathrm{tr}(\rho_1(g)) + \mathrm{tr}(\rho_2(g)) = \chi_1(g) + \chi_2(g), \quad g \in G.$$

The **conjugacy class** of an element $a \in G$ is the set $\mathrm{Cl}(a) = \{gag^{-1} \mid g \in G\}$. The character of any representation is constant on conjugacy classes! In other words, fixing $a \in G$, we have $\chi_\rho(b) = \chi_\rho(a)$ for any $b \in \mathrm{Cl}(a)$ and any representation ρ of G .

More q -Series

Assign a vector space $V^{(n)}$ to each exponent n of q in the function j that is a direct sum of irreps. V_i of $\mathcal{M}$. The **Monster rep.** is $\mathcal{V} = \bigoplus_{n=-1}^{\infty} V^{(n)}$. Also,

$$V^{(-1)} = V_1, \quad \text{so } \dim(V_1) = 1,$$

$$V^{(0)} = 0,$$

$$V^{(1)} = V_1 \oplus V_2, \quad \text{so } \dim(V_2) = 196883,$$

$$V^{(2)} = V_1 \oplus V_2 \oplus V_3, \quad \text{so } \dim(V_3) = 21296876.$$

A **McKay–Thompson** series associated to $\mathcal{V}$ is defined for a $g \in \mathcal{M}$ via:

$$T_g(\tau) = \sum_{n=-1}^{\infty} \text{tr}_{V^{(n)}}(g) q_2^n = q_2^{-1} + 0 + [\chi_1(g) + \chi_2(g)] q_2 + \dots, \quad \tau \in \mathbb{C}_+,$$

where, $q_2 = e^{2\pi i \tau} = q^2$ and $\chi_i(g) = \text{tr}(\rho_i(g)) = \text{tr}_{V_i}(g)$. We quickly find that

$$T_e(\tau) = \sum_{n \geq -1} \dim(V^{(n)}) q_2^n = j(\tau) - 744 \quad \tau \in \mathbb{C}_+$$

Each element in the same conjugacy class has the same McKay-Thompson series.

The Main Result

Monstrous Moonshine Conjecture (J. Conway, S. Norton 1979)

For each $g \in \mathcal{M}$ the McKay-Thompson series T_g coincides with the unique Hauptmodul J_{Γ_g} with expansion $q_2^{-1} + O(q)$ near $\tau \rightarrow i\infty$ for some genus zero subgroup $\Gamma_g \leq \mathrm{SL}(2, \mathbb{R})$. Furthermore, each Γ_g contains $\Gamma_0(N)$ as a normal subgroup for some N dividing the quantity $|g| \gcd(24, |g|)$, where $|g|$ is the integer such that $g^{|g|} = e$ in $\mathcal{M}$.

- Hauptmodul: Normalized (no q_2^0 term) generator of the space of functions automorphic with respect to Γ_g . $j(\tau) - 744$ is the $\Gamma_e = \mathrm{SL}(2, \mathbb{Z})$ Hauptmodul i.e. any function invariant under Γ_g is a rational function of the Hauptmodul
- Γ_g is a genus zero subgroup if its fundamental domain is a genus zero surface
- $\Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}(2, \mathbb{Z}) \mid c \equiv 0 \pmod{N} \right\}$ is of genus 0 for $N < 10$
- $\Gamma_0(N)^+$ is the group $\Gamma_0(N)$ including the generator $\frac{1}{\sqrt{N}} \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}$
- A. Ogg (1974): $\Gamma_0(p)^+$ is genus zero iff p is a prime which divides $|\mathcal{M}|$

Examples of Hauptmoduln (plural for Hauptmodul) with $E_2 = Es_1$ and $\eta(\tau)$ is expanded in q_2 rather than q .

N	1	2	3	4	5	6
$J^{(N)}(\tau)$	$\frac{E_4(\tau)^3}{\eta(\tau)^{24}} - 744$	$\frac{\eta(\tau)^{24}}{\eta(2\tau)^{24}} + 24$	$\frac{\eta(\tau)^{12}}{\eta(3\tau)^{12}} + 12$	$\frac{\eta(\tau)^8}{\eta(4\tau)^8} + 8$	$\frac{\eta(\tau)^6}{\eta(5\tau)^6} + 6$	$\frac{\eta(\tau)^5\eta(3\tau)}{\eta(2\tau)\eta(6\tau)^5} + 5$

N	7	8	10	12	16
$J^{(N)}(\tau)$	$\frac{\eta(\tau)^4}{\eta(7\tau)^4} + 4$	$\frac{\eta(\tau)^4\eta(4\tau)^2}{\eta(2\tau)^2\eta(8\tau)^4} + 4$	$\frac{\eta(\tau)^3\eta(5\tau)}{\eta(2\tau)\eta(10\tau)^3} + 3$	$\frac{\eta(\tau)^3\eta(4\tau)\eta(6\tau)^2}{\eta(2\tau)^2\eta(3\tau)\eta(12\tau)^3} + 3$	$\frac{\eta(\tau)^2\eta(8\tau)}{\eta(2\tau)\eta(16\tau)^2} + 2$

TABLE 5.1. Hauptmoduln for some $\Gamma_0(N)$

N	11	14
$J^{(N,+)}(\tau)$	$-\frac{E_2(\tau)-11E_2(11\tau)}{10\eta(\tau)^2\eta(11\tau)^2} - \frac{22}{5}$	$-\frac{E_2(\tau)+2E_2(2\tau)-7E_2(7\tau)-14E_2(14\tau)}{18\eta(\tau)\eta(2\tau)\eta(7\tau)\eta(14\tau)} - \frac{7}{3}$

N	15	16
$J^{(N,+)}(\tau)$	$-\frac{E_2(\tau)+3E_2(3\tau)-5E_2(5\tau)-15E_2(15\tau)}{16\eta(\tau)\eta(3\tau)\eta(5\tau)\eta(15\tau)} - \frac{5}{2}$	$\frac{\eta(2\tau)^6\eta(8\tau)^6}{\eta(\tau)^4\eta(4\tau)^4\eta(16\tau)^4} - 4$

N	19	20	28
$J^{(N,+)}(\tau)$	$-\frac{E_2(\tau)-19E_2(19\tau)}{18f_{19}(\tau)} - \frac{4}{3}$	$\frac{\eta(2\tau)^8\eta(10\tau)^8}{\eta(\tau)^4\eta(4\tau)^4\eta(5\tau)^4\eta(20\tau)^4} - 4$	$\frac{\eta(2\tau)^6\eta(14\tau)^6}{\eta(\tau)^3\eta(4\tau)^3\eta(7\tau)^3\eta(28\tau)^3} - 3$

N	31	32
$J^{(N,+)}(\tau)$	$\frac{\sqrt{5}(f_{31}(\tau)+f_{31}^{\sigma}(\tau))}{2(f_{31}(\tau)-f_{31}^{\sigma}(\tau))} - \frac{5}{2}$	$\frac{\eta(2\tau)^3\eta(16\tau)^3}{\eta(\tau)^2\eta(4\tau)\eta(8\tau)\eta(32\tau)^2} - 2$

TABLE 5.2. Hauptmoduln for some $\Gamma_0^+(N)$